Applying finishing touches Activity Book

Data Modeling for Qlik Sense

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Activity 1: Use set analysis in a table

Summary:

- Create a product-focused table object to analyze sales.
- Leverage set analysis syntax within the table measures.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, use set analysis syntax to create a table that compares the following:

- *Total Sales* by *ProductName*
- *Sales* for *USA* only
- *Sales* for all countries except *USA*
- Percentage of *USA sales* over *Total sales*

Be prepared to show these comparisons by completing the following tasks:

- Create a new sheet named *Advanced Calculations* and insert a **Table** object with *ProductName* as the dimension. Change the label to *Products*.
- Add three measure columns to calculate the following:
  - *All Sales* (use the *LineSalesAmount* field)
  - *Sales USA* (use the *LineSalesAmount* to calculate only the *USA sales* with the use of set analysis)
  - *Sales Non USA* (use the *LineSalesAmount* field to calculate all **but** the *USA sales*)
  - *%USA Sales* (`[Sales USA] / [All Sales]`)
- Set the number format to *Money* for all the above.

Step-by-step guide

1. Using **Analyze** mode, create a new sheet and name it *Advanced Calculations*.
2. In sheet editing mode, insert a **Table** object onto the top half of the new sheet.
3. Add the *ProductName* field to the table as a dimension and label it *Products*.
4. Add the *LineSalesAmount* field as a measure and select the *Sum(LineSalesAmount)* aggregation.
5. Change its label to *All Sales* and set the number format to *Money*.

   Note: When using **Money** as the **Number formatting**, Qlik Sense utilizes the settings from the data load script to determine the format pattern. If required, it can be overridden.

6. Add the *LineSalesAmount* field as a measure for a second time, and select the *Sum (LineSalesAmount)* aggregation again.
7. This time, label the measure *Sales USA* and set the number format to *Money*.
8. Apply set analysis in the new expression as follows:
   `Sum({$<Country={USA}>} LineSalesAmount)`

   Note: Since the value *USA* does not include any spaces, the use of quotation marks is optional.

9. Add a third measure for sales, for all countries except *USA*: Use the *LineSalesAmount* field as a measure, with the same aggregation, *Sum(LineSalesAmount)*.
10. Change the measure label to *Sales Non USA* and set the number format to *Money*.
11. Edit the expression using set analysis again (notice the minus sign):
    `Sum({$<Country=-{USA}>} LineSalesAmount)`
12. Add one final new measure column in the table with the following expression:
    `[Sales USA] / [All Sales]`

13. Label the last measure *%USA Sales* and set the number format to *Number*, selecting the *12.3%* option.
14. Stop editing the sheet. Verify that your table contains three sales columns where *All Sales* is equivalent to the other two columns added together. Confirm that a fourth column contains the percentage of the second column over the first column.

Solution

Advanced Calculations

Products	All Sales	Sales USA	Sales Non USA	%USA Sales
Totals	**$12,233,524.28**	**$1,569,156.50**	**$10,664,367.78**	**12.8%**
AAA Running Shoe	$137,065.12	$16,101.76	$120,963.36	11.7%
Aino Shoes	$311,000.00	$31,187.50	$279,812.50	10.0%
Atles Lussekofta	$194,605.25	$25,482.75	$169,122.50	13.1%
Baby Dark Lounge Suit	$10,111.34	$924.27	$9,187.07	9.1%
Basket Shoes	$6,500.50	$829.00	$5,671.50	12.8%
Basket Vest	$3,439.80	$504.70	$2,935.10	14.7%
Baywatch Swimsuit	$27,513.60	$4,372.80	$23,140.80	15.9%
Bike Helmet	$16,264.58	$2,695.15	$13,569.43	16.6%
Bow tie	$5,275.81	$611.09	$4,664.72	11.6%
Cap	$6,715.00	$873.75	$5,841.25	13.0%
Casual Boots	$355,788.75	$42,234.37	$313,554.37	11.9%
Chantell Shirt	$95,256.00	$16,200.00	$79,056.00	17.0%
Danske Treshoe	$217,835.00	$28,595.00	$189,240.00	13.1%
Davenport Shoes	$50,481.56	$5,985.25	$44,496.32	11.9%
Desperado Jeans	$356,090.00	$53,340.00	$302,750.00	15.0%

Activity 2: Set analysis filtering with a field

Summary:

- Develop a Line chart which uses calendar flags inside set analysis as measures.
- Develop a Bar chart which utilizes set analysis and a calendar flag to analyze year-to-date sales.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, create the following two visualizations:

- A **Line** trend showing current versus last year-to-date *Sales per Month*.
- A **Bar chart** of *Sales per Year* in the same period (year-to-date).

To achieve that, do the following:

- On the bottom left of the **Advanced Calculations** sheet, insert a **Line chart** with *Month* as the dimension and the following measures:
  - *Cur YTD Sales*, with an expression that calculates the total sales (*LineSalesAmount*) where *CYTDFlag=1* (use set analysis to do that).
  - *Last YTD Sales*, with a similar expression that calculates the total sales where *LYTDFlag=1*.
- In the remaining space, insert a **Bar chart** onto the sheet, with *Year* as the dimension and *YTD Sales* as a the measure that calculates the sales where *IsInYTD=1*.
- Make sure the bar chart is sorted by *Year*.

Hints

- The *MasterCalendar* table contains flags: *CYTDFlag*, *LYTDFlag*, and *IsInYTD* that are set to *1* if the date is in the year-to-date period. These flags can be used in the set analysis expressions to simplify the calculations.

Step-by-step guide

1. In the **Advanced Calculations** sheet, insert a **Line chart** object onto the bottom left area.
2. Add the *Month* field as the dimension and the following expression as the measure:
   `Sum({$<CYTDFlag={1}>} LineSalesAmount)`

   Note: The use of quotation marks for numerical set modifier values is optional.
3. Label the measure *Cur YTD Sales*.
4. Add a second measure to the chart, *Last YTD Sales*, using the following expression:
   `Sum({$<LYTDFlag={1}>} LineSalesAmount)`
5. From the **Assets** panel, insert a **Bar chart** object onto the bottom right area of the sheet.
6. Use the *Year* field as the dimension and the following expression as the measure:
   `Sum({$<IsInYTD={1}>} LineSalesAmount)`
7. Label the measure *YTD Sales* and make sure that the bar chart is sorted by *Year*.

   Note: These expressions use *CYTDFlag*, *LYTDFlag*, and *IsInYTD* fields calculated in the script to identify the dates in the year-to-date period. The expressions would be more complex without these fields.
8. Disable the edit mode and analyze the bar chart. The value for each year is only calculated for the yearly period that matches the year-to-date period for the current year at the time of the last data load. This is possible using the *IsInYTD* field, calculated in the *MasterCalendar* table.

Solution

Advanced Calculations

Products	All Sales	Sales USA	Sales Non USA	%USA Sales
Totals	**$12,233,524.28**	**$1,569,156.50**	**$10,664,367.78**	**12.8%**
AAA Running Shoe	$137,065.12	$16,101.76	$120,963.36	11.7%
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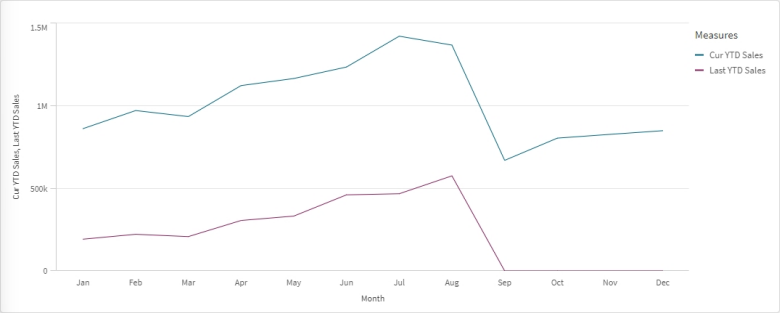
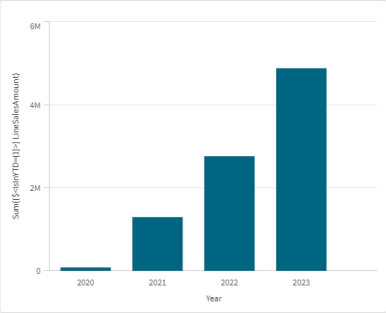




Activity 3 - Optional: Correct set analysis syntax

Summary:

- Analyze numerous set analysis expressions.
- Identify possible errors and use cases.

Qlik Sense app: ABC Sales Data

Challenge

Review the expressions using set analysis, identify possible errors, and suggest corrections.

Note that these expressions are using the data fields and values from the **ABC Sales Data** app.

Hints

- Review the table below and check whether the set analysis syntax in these expressions could be used in the app.
- Consider the answers to these questions as they relate to each expression:
- Is the expression correct if it is used in the app? If not, could you correct it so it is a correct expression for the app?
- Please be aware that not all the expressions have syntax errors.

ID	Expression
1	`Sum({1<country={USA}>} Sales)`
2	`Count({<Year={2021}, Month=>}Distinct OrderID)`
3	`Avg({$<OrderDate.autoCalendar.InYTD={1}>} Sales)`
4	`Count(${<Division={America}> distinct Customer)`
5	`Sum({$<Division=EMEA, APAC>} Sales)`
6	`Sum({<Month={"<3"}>} Margin)`
7	`Sum({$<Product category={“Footwear”}+{“Baby”}>} Sales)`

Activity 4: Use the Aggr function in a reference line

Summary:

- Add a reference line to a bar chart to indicate average sales.
- Develop and test the aggregation function, in conjunction with set analysis syntax, to achieve the desired results.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, use the **Aggr** function to add a reference line to the *YTD Sales per Year* bar chart. The reference line shows the average *YTD Sales*, and can be created by doing the following:

- Open the **Advanced Calculations** sheet. Add a reference line to the *YTD Sales per Year* bar chart with the following characteristics:
  - *Avg Sales* as the label.
  - *#e1201d* as the color.
  - `Avg(Aggr(Sum(LineSalesAmount), Year))` as an expression.
- Modify the reference line to only show the *Current Year Avg Sales* (*IsInYTD = {1}*).
- Modify the reference line expression, again, to make it ignore all the selections on the *Year* field.

Hints

- The expression for the reference line is modified to ignore time-related selections. Otherwise, the line might disappear from the chart area when there are selections in the app.

Step-by-step guide

1. In the **Advanced Calculations** sheet, enable the edit mode.
2. Select the bar chart and locate the **Reference lines** option within the **Add-ons** section of the **Properties** panel.
3. Add a reference line to the chart and set the label to *Avg Sales* and the color to *#e1201d*.
4. Edit the **Reference line expression** with the following:
   ```
   Avg(Aggr(Sum(LineSalesAmount), Year))
   ```
   The result of the **Aggr** function is a set of total sales values for each *Year*. The **Avg** aggregation function is then used to find the average of the set of sales values.
5. Disable the edit mode and review the bar chart. Can you see the reference line?
   Now make select a single year value and confirm the selection.
   Notice how a **1** has appeared to the top right side of the chart. This line does not show the average of the visible bars in the chart. Why?
6. Enable the edit mode again. Select the bar chart and expand the **Reference lines** in the **Properties** panel.
7. Modify the **Reference line expression** for the *Avg Sales* as follows:
   ```
   Avg(Aggr(Sum({$<IsInYTD={1}>} LineSalesAmount), Year))
   ```
8. Disable the edit mode and review the bar chart again, making selections on the *Year* field. This time, the *Avg Sales* reference line is displayed in the bar chart with the value.
9. Ensure the selection state represents the very final year for which there is data, and confirm the selection. Has the reference line changed? Why?
10. If you want the *Avg Sales* reference line to stay as the average of all the *Years*, even when you select one or more values in the *Year* field, you need to change the expression in the reference line as follows:
    ```
    Avg({$<Year=>} Aggr(Sum({$<IsInYTD={1}, Year= >} LineSalesAmount),
    Year))
    ```
    Notice how the set expression in the original **Sum** function has changed to include *Year=*. Also, the **Avg** function now includes a set expression to avoid the selections in the *Year* field affecting the calculation for the reference line.
11. Enable the edit mode again and change the **Reference line expression** as described in the previous step.
12. Finish editing the sheet and select a different *Year*. What happens with the reference line? What happens if you select a product from the table in the same sheet?

> **Note:** Knowing your data is very important when you work in either the front-end or the back-end of a Qlik Sense app. In this data set, we know that there are customers who have never placed orders and products that have never been sold. Because of the default **Join** type that Qlik Sense uses (**Outer Join**), associations are created for the products and the customers with zero sales in all tables, and populates them with null values. Cases like this often result in an extra dimension value (null). In this specific case, this dimension has zero sales.
>
> We can easily verify that if we create a table and use two dimensions: *Year* and the following **Aggr** expression:
> ```
> Aggr(Sum({$<IsInYTD={1}, Year= >} LineSalesAmount), Year)
> ```
>
> This returns a list of the sales values we see as bars, per each year, but it also includes a null year (-) with zero sales.

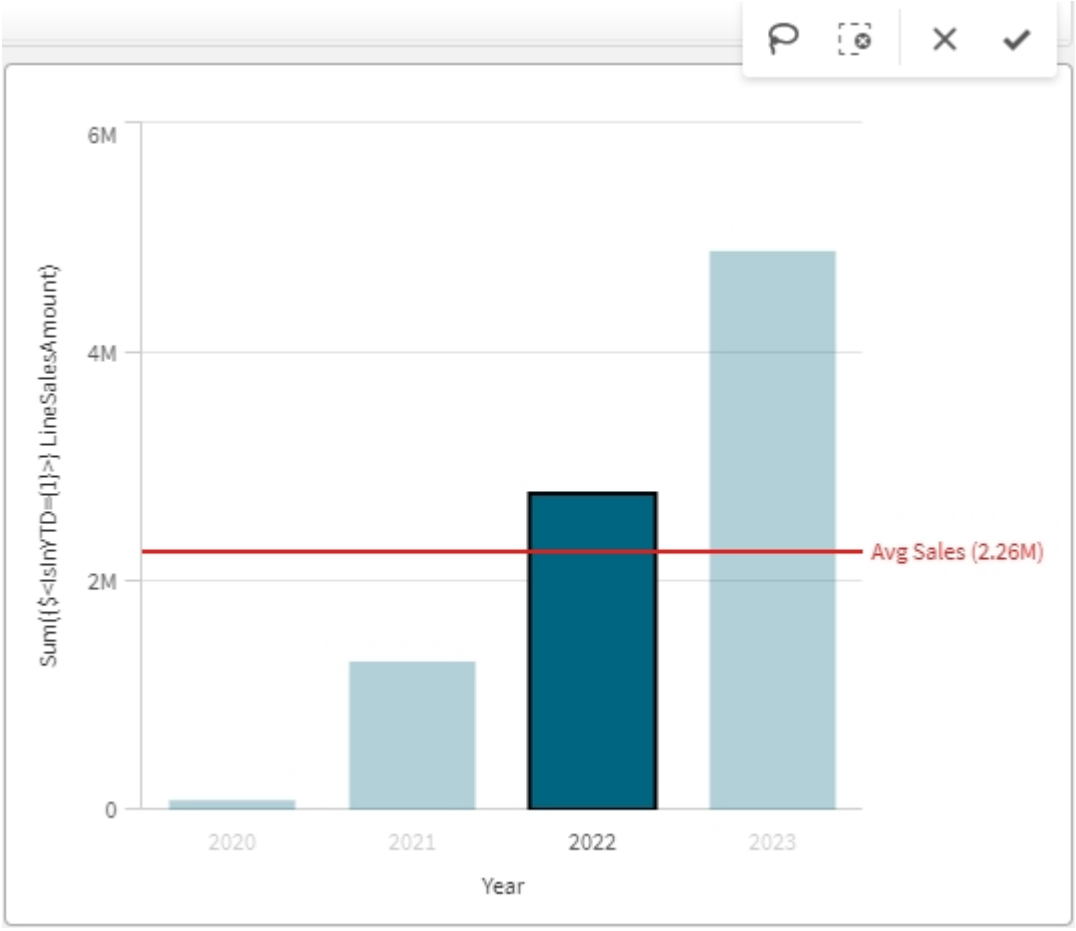
The **Avg** function calculates the zero and returns the mean of all values, which in this case is invalid.

To avoid this, reduce the scope of the expression and only apply data where the associated sales is more than zero. The above expression will become:

```
Avg({$<Year=>} Aggr(Sum({$<LineSalesAmount={">0"}, IsInYTD={1},
Year= >} LineSalesAmount), Year))
```

13. Clear all your selections once you finish the analysis.

Solution



Activity 5: TOTAL versus ALL versus {1}

Summary:

- Develop a bar chart with DivisionName as the dimension and various LineSalesAmount measures.
- Analyze differences in the measure columns when using various qualifier prefixes.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, create a **Bar chart** to better demonstrate the difference between **Total**, **All** and **{1}** qualifiers by doing the following:

- Create a new sheet named *Qualifiers* and add the *DivisionName* field as a **Filter pane** on the top of the sheet.
- Insert a **Bar chart** with *Year* as the dimension and *Sum(LineSalesAmount)* as the measure.
- Modify the above measure expression to *Sum(TOTAL LineSalesAmount)* and observe the results.
- Create a new measure on the same chart with the **All** qualifier: *Sum(ALL LineSalesAmount)*
- Select a value in the *DivisionName* field and observe the different behavior.
- Create a third measure using the **{1}** set analysis identifier: *Sum({1} LineSalesAmount)*
- Make selections and observe the differences.

Hints

- The first measure uses the **Total** qualifier. This expression takes into account the user selections, but ignores the dimension values.
- The second measure uses the **All** qualifier. This expression will ignore both the dimension values and the user selections.
- The third measure uses the set analysis identifier **{1}**. This expression aggregates the data by dimension values but ignores the user selections.

Step-by-step guide

1. Using **Analyze** mode, create a new sheet named *Qualifiers*.
2. In the new sheet, clear the selections, if any, and enable the edit mode.
3. From the **Assets** panel, drag the *DivisionName* field to the top row. Make sure you use the **Shift** key when you drag the field to add it as a **Filter pane**.
4. Insert a **Bar chart** onto the sheet with *Year* as the dimension and *Sum(LineSalesAmount)* as the measure.
5. Label the measure *TOTAL Sales* and modify the expression as follows:
   `Sum(TOTAL LineSalesAmount)`
6. Disable the edit mode. From the *DivisionName* filter pane, select *North America* and confirm your selection.

   NOTE: After modifying the expression, you see that all the bars in your bar chart have the same height. This is because the **TOTAL** qualifier forces the expression to ignore the

dimensional values of the chart.
Selecting a *DivisionName* value affects the *TOTAL Sales* expression and changes the height of the bars. However, the dimensional values are still being ignored. As a result, all the bars still have the same height.

7. Clear your selections and enable the edit mode again for the sheet. Add a new measure to the bar chart, using the *LineSalesAmount* and the *Sum(LineSalesAmount)* aggregation option, just like before.
8. Label the new measure *ALL Sales* and modify the expression as follows:
   `Sum(ALL LineSalesAmount)`
9. Disable the edit mode and from the *DivisionName* filter pane, select *North America* again and confirm your selection.

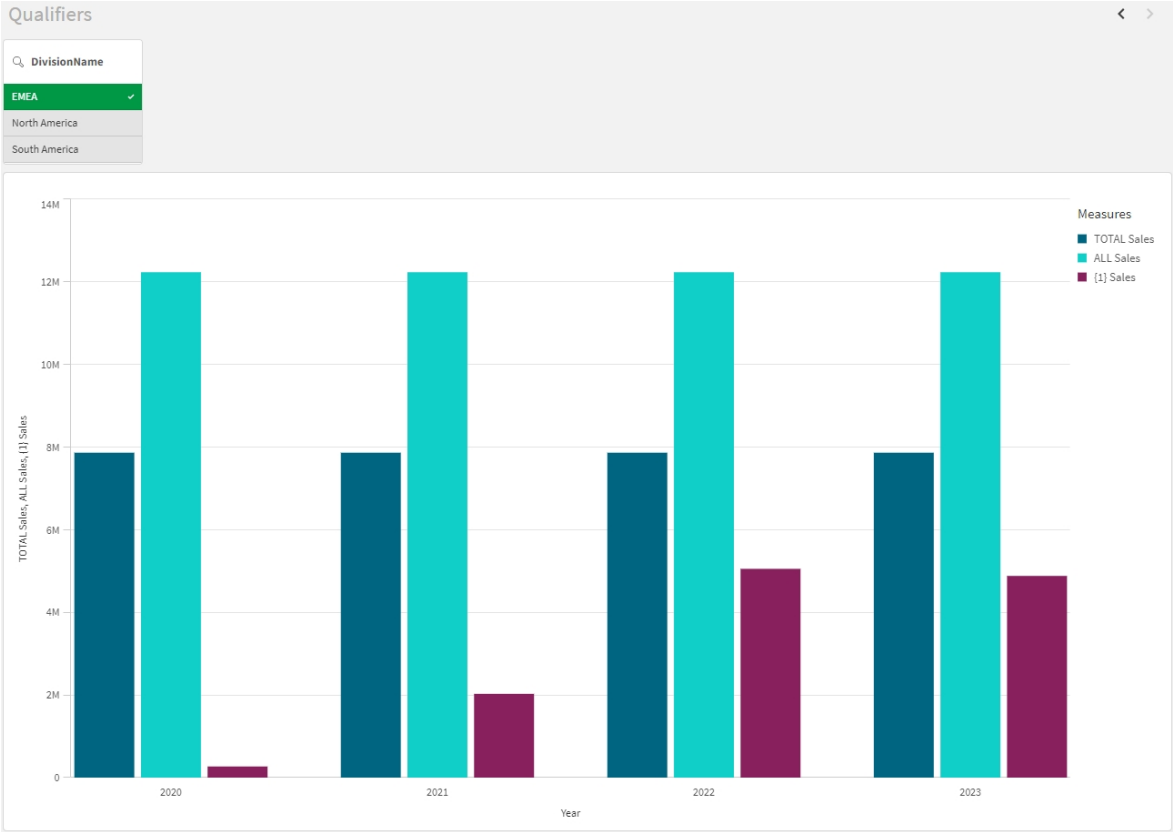
NOTE: After modifying the expression, you see that all the bars in your bar chart have the same height. This is because the **ALL** qualifier forces the expression to ignore the dimensional values of the chart.
Selecting a *DivisionName* value does not affect the *ALL Sales* expression. The height of the bars are unchanged even after making selections. This is because the **ALL** qualifier also ignores the user selections.

10. Clear your selections and enable the edit mode for a third time. Add a third measure, using the same aggregation, *Sum(LineSalesAmount)*.
11. Label the third measure *{1} Sales* and modify the expression as follows:
    `Sum({1} LineSalesAmount)`
12. Disable the edit mode. From the *DivisionName* filter pane, select *EMEA* and confirm your selection.

NOTE: After modifying the expression, you see that a new set of bars appears in your bar chart. These bars take into account the dimensional value for each bar, showing correct results.
Selecting a *DivisionName* value does not affect the *{1} Sales* expression. The height of the bars are unchanged even after making selections. This is because the **{1}** set analysis identifier always disregards the user selections.

13. Clear all your selections.

Solution



Activity 6: Master Dimensions from fields

Summary:

- Create Master Dimension items for the application, via the Data model viewer.
- Observe the items from the edit sheet view perspective.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, create three Master Dimensions by doing the following:

- In your data model, preview the *MasterCalendar* table and select the *Month* field.
- Use the **Preview** panel to add the *Month* field as a Master Dimension.
- Add tags and a brief description for *Month* before you create the Master item.
- Create two more Master Dimensions, *Category* from the *CategoryName* field and *Product* from *ProductName*.

- In the **App overview**, open the first sheet. In edit mode, use the **Assets** panel to locate and verify that the Master Dimensions have been added.

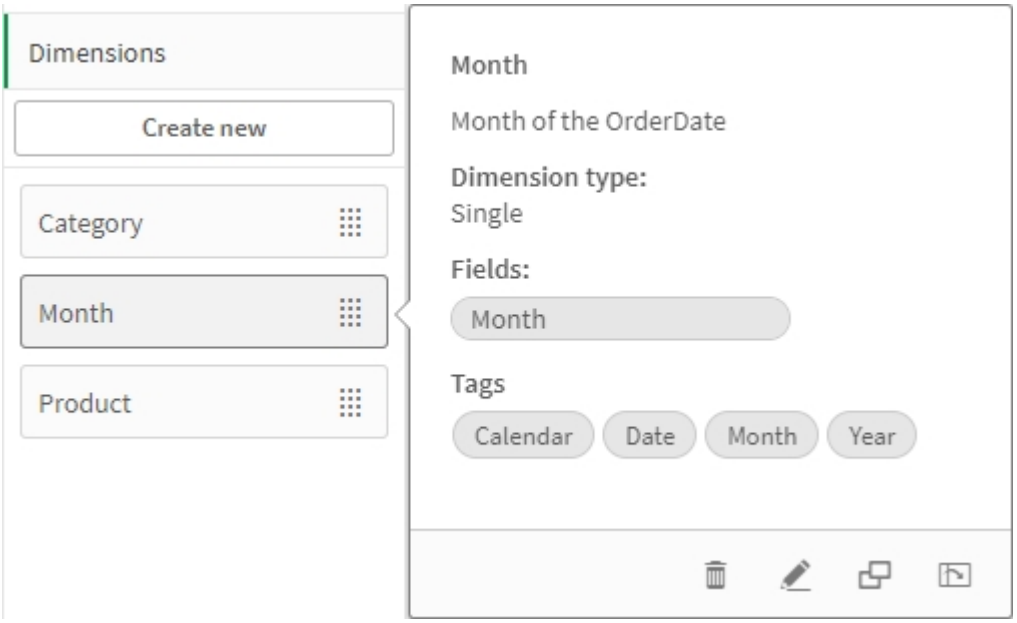
Hints

- The Master items can be reused multiple times within an app.
- Adding tags to the Master items makes them more searchable.

Step-by-step guide

1. In your data model, preview the *MasterCalendar* table and select the *Month* field.
2. Use the **Add as dimension** button in the **Preview** panel to create a Master Dimension.
3. Ensure the **Name** of the dimension is set to *Month* and enter *Month of the OrderDate* as **Description**.
4. In the **Tags** input box, add the following tags: *Calendar*, *Date*, *Month*, and *Year*.
5. Create the dimension. Look for the message to verify that the dimension has been created.
6. While the **Create new dimension** window is still open, add a second Master Dimension for the *CategoryName* field. Set the **Name** to *Category* and the **Description** to *Product category*.
7. Add the following two **Tags**: *Product* and *Category*. Then, click the **Create** button.
8. Add a third Master Dimension for the *ProductName* field. Set the **Name** to *Product* and the **Description** to *Product items from catalogue.*
9. Add the following three **Tags**: *Category*, *Item*, and *Supplier*. Then, click the **Create** button.
10. Close the **Create new dimension** window and navigate to **Analyze** mode to open the first sheet.
11. Enable the sheet edit mode and locate the **Master items** tab in the **Assets** panel.
12. Open the **Dimensions** section to display the list of Master Dimensions. Then, click *Month*. A pop-up is displayed showing the attributes for this item.
13. Once you finish observing the **Master Items** library, stop editing the sheet.

Solution



Activity 7 - Optional: Master Dimension from an expression

Summary:

- Use the Data model viewer to generate a new Master Dimension, based on an expression of Quarter and Year.
- Use the new Master Dimension to replace the existing dimension on a Line chart.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, create a calculated dimension and add it to the **Master items** library by doing the following:

- In the **Data model viewer**, preview the *Quarter* field in the *LinkTable* and select the button to **Add as dimension**.
- Use the expression =*Quarter & '-' & Year* and name the new Master Dimension *Quarter Year*. Add **Tags** and a **Description**.
- Navigate to the **Sales vs. Quota** sheet and add the new Master Dimension, *Quarter Year*, as a **Filter pane**.
- On the line chart, replace the *Year* dimension with *Quarter Year*.
- Sort the line chart by expression. Use (*Year * 10 + Right(Quarter, 1)*) to fix the sorting issue.

Hints

- The *Quarter Year* Master Dimension is based on the *Quarter* and *Year* fields.
- Once the *Quarter Year* Master Dimension is created, it can be reused multiple times in the app.

Step-by-step guide

1. In the **Data model viewer**, select the *Quarter* field in the *LinkTable* and preview the table.
2. Select the button to **Add as dimension**.
3. In **Field**, replace the current field with the following expression:
   `=Quarter & '-' & Year`
4. **Name** the Master Dimension *Quarter Year* and set the **Description** to *Quarter Year field displayed as QQ-YYYY*.
5. Add the following four **Tags**: *Calendar*, *Date*, *Month*, and *Year*.
6. Click the **create** button, close the window,y and navigate into **Analyze** mode to open the **Sales vs. Quota** sheet.
7. Enable the edit mode. Add the *Quarter Year* Master Dimension from the **Assets** panel, as a **Filter pane** on the sheet. Notice the data displayed in the format *QQ-YYYY*.

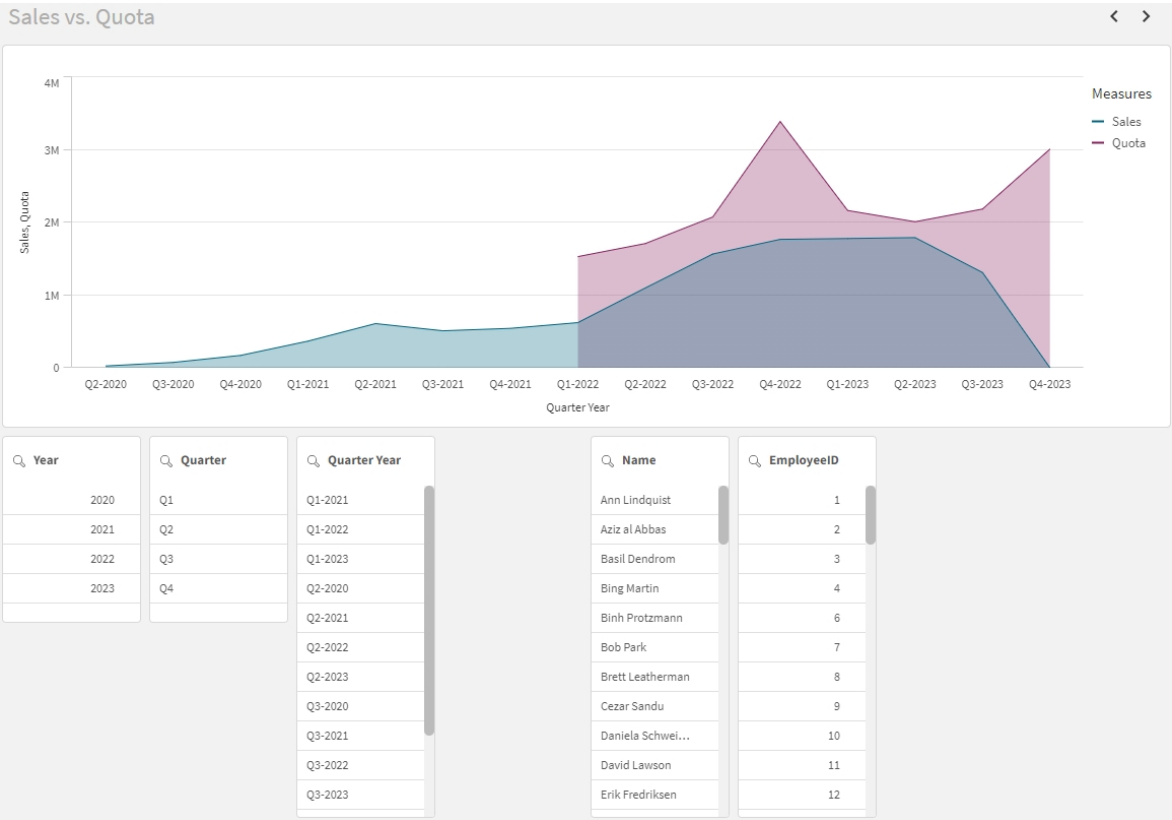
8. Drag the *Quarter Year* Master Dimension on the line chart to replace the existing *Year* field. The chart dimension changes to *Quarter Year*.
9. Review the line chart. Is there any problem with the new dimension? The order of the values is not correct in the time trend. You need to resolve the sorting issue.
10. In the **Properties** panel, modify the **Sorting** options for the line chart as follows:
    a. Sort the chart by *Quarter Year.*
    b. Disable the **Automatic** sorting.
    c. Sort the chart by the following expression, ascending:

    ```
    Year * 10 + Right(Quarter, 1)
    ```

> **Note:** As there is no easy way to sort the new *Quarter Year* field, you have to **Sort by expression** and use the above formula. This formula multiplies the *Year* value by 10 and adds the *Quarter* number to it. So, the list of values it returns and uses for sorting is: *...20153, 20154, 20161, 20162...*

11. Stop editing the sheet and observe the results.

Solution



Activity 8: Create a Drill-down Master Dimension

Summary:

- Develop a Drill-down Master Dimension using date-related fields.
- Use the Drill-down Master Dimension on a new table.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, create a Drill-down Master Dimension by doing the following:

- From the **Data model viewer**, preview the *Year* field from the *LinkTable* and **Add as dimension**.
- Choose **Drill-down** and add the fields *Quarter, Month*, and *Day* from the list.
- Use *Year Drill-down* as the dimension **Name**. Add some **Tags** and a brief **Description**.
- Create a new app sheet, *Master Items*. Insert a **Table** object with *Year Drill-down* as the dimension and *Sum(LineSalesAmount)* as the measure.
- Change the **Number formatting** of the measure to *Money*.
- Make selections to see how the Drill-down Master Dimension works. After you are finished, clear all selections.

Hints

- The order with which the fields are added to a Drill-down Master Dimension is significant. The higher level dimensions need to be added first on the list.
- The new Master Dimension is based on the hierarchy for *Year*, *Quarter*, *Month*, and *Day* fields.

Step-by-step guide

1. In the **Data model viewer**, preview the *Year* field from the *LinkTable* and select the button to **Add as dimension**.
2. Select the **Drill-down** option from the **Create new dimensions** window.
3. *Year* is already added as a field. Search for and add the fields *Quarter, Month*, and *Day* from the field list, and notice how they are included in the box on the right. Each field is a level in the drill-down hierarchy. It is important to have the highest level of the dimensions at the top of the list.
4. Use *Year Drill-down* as the dimension **Name**, and add the **Description** *Year-Quarter-Month-Day drill-down dimension*.
5. Add the following six **Tags**: *Calendar*, *Hierarchy*, *Day*, *Month*, *Quarter*, and *Year*.

   Note: You might need to add the tags in two stages, as the number of maximum input characters in the **Tags** field is limited to *31*.

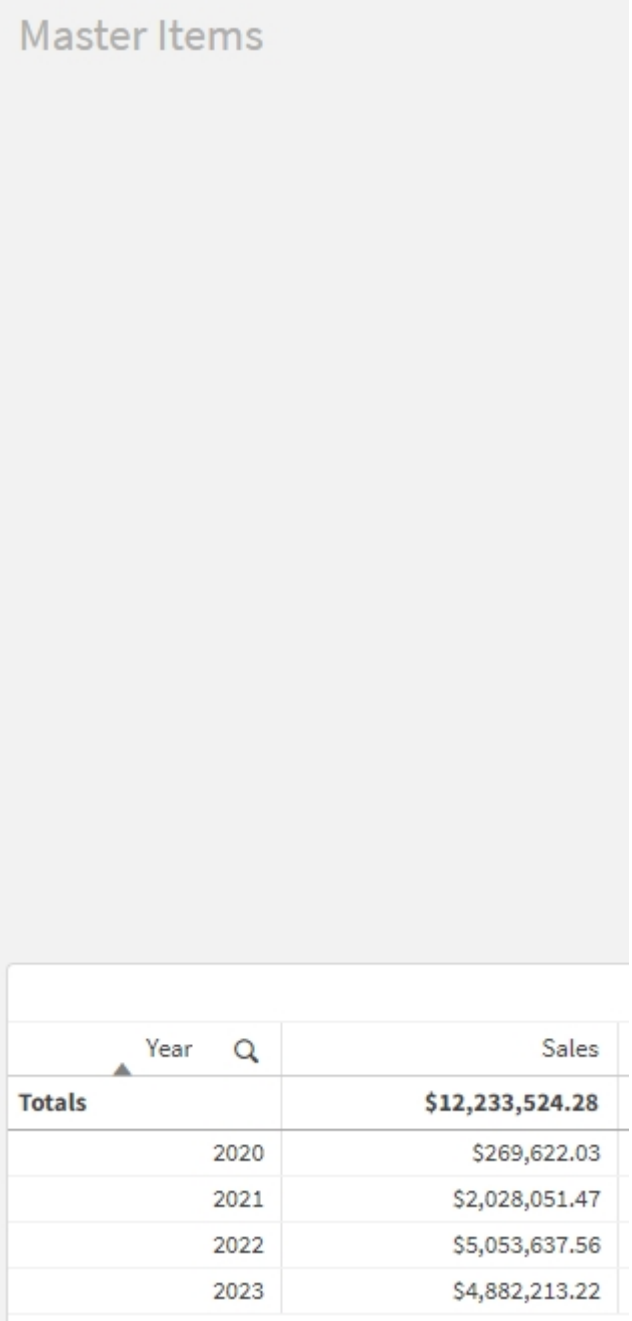
6. Click the **Create** button, close the window, and navigate to **Analyze** mode to create a new sheet.
7. Name the new sheet *Master Items* and insert a **Table** object onto its bottom half. In the **Master items** area, locate the *Year Drill-down* Master Dimension and add it to the table.

8. Add the *LineSalesAmount* field as the measure to the table, using the *Sum* aggregation.
9. Name the measure *Sales* and change the **Number formatting** to *Money*.

> **Note:** When using **Money** as the **Number formatting**, Qlik Sense utilizes the settings from the data load script to determine the format pattern. If required, it can be overridden.

10. Disable the edit mode. In the table's *Year* column, select *2023*. Notice that when you confirm your selection, the column drills down to *Quarter*. Select *Q3* and confirm your selection again. Notice, the column drills down to *Month*.
11. Clear all the selections to return to the *Year* dimension.

Solution



Year	Sales
Totals	**$12,233,524.28**
2020	$269,622.03
2021	$2,028,051.47
2022	$5,053,637.56
2023	$4,882,213.22

Activity 9: Create Master Measures

Summary:

- Create Master Measures for use in analysis mode.
- Leverage the Master Measures in a table object.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, create two Master Measures using simple aggregation functions. To achieve this, complete the following tasks:

- From the **Data model viewer**, preview the *LineSalesAmount* field and **Add as measure**.
- Replace the **Expression** with*Sum(LineSalesAmount)*. Add some **Tags** and a brief **Description**. Use *Sales* as the new measure **Name**.
- Add a second Master Measure named *COGS,* with *Sum(CostOfGoodsSold)* as the **Expression**.
- In the **Master Items** sheet, add the new Master Measure *Sales* and replace the existing *Sales* column.
- Add *COGS* as a column in the same table.
- For both Master Measures, change the **Number formatting** to *Money*.

Hints

- Once a Master Measure is created, it can be reused multiple times in the app.
- Modifying the expression of a Master Measure will update all the objects that use this measure.

Step-by-step guide

1. In the **Data model viewer**, select the *LineSalesAmount* field from the *OrderDetails* table.
2. Preview the table and select the button to **Add as measure**.
3. Change the Master Measure **Expression** to the following:
   `Sum(LineSalesAmount)`
4. Set the **Name** to *Sales* and add the following **Description**: *Sales value at order line level*.
5. Set the **Number formatting** to *Money*.

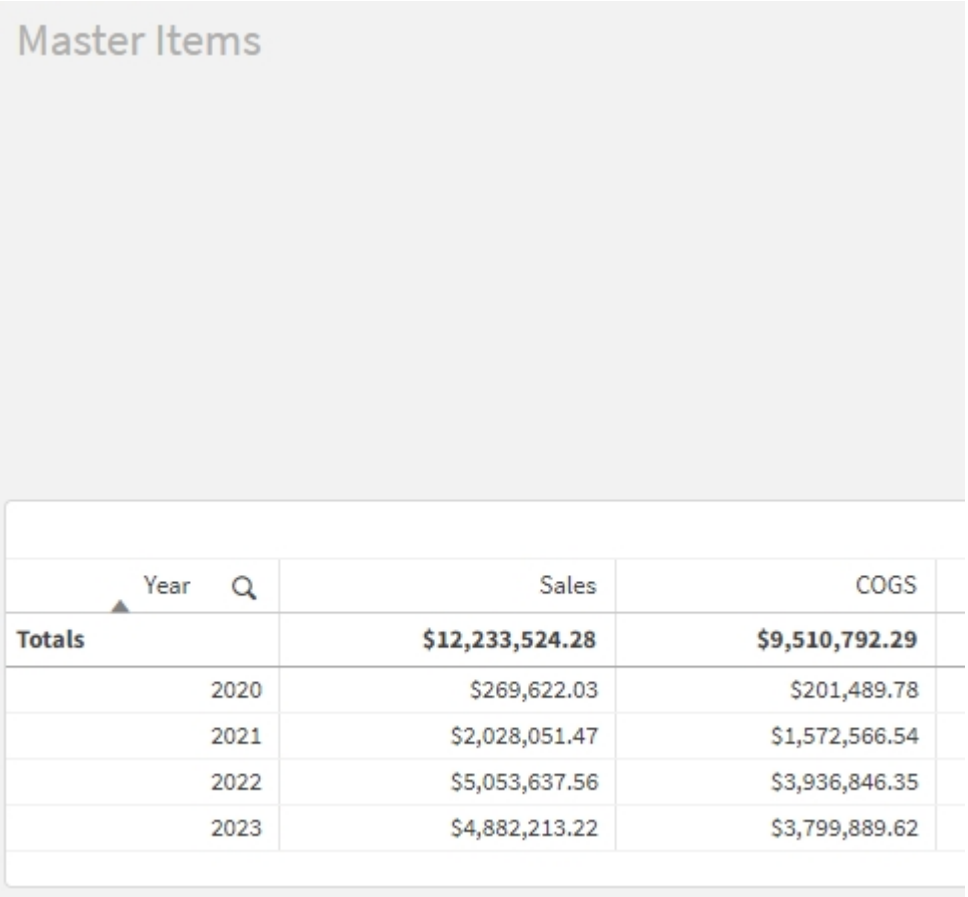
   Note: When using **Money** as the **Number formatting**, Qlik Sense utilizes the settings from the data load script to determine the format pattern. If required, it can be overridden.

6. Add four **Tags**: *Amount*, *Line*, *Order*, and *Sales*.

7. **Create** the Master Measure.
8. Repeat the process for the *CostOfGoodsSold* field, from the *OrderDetails* table: Select the button to **Add as measure** and change the **Expression** to the following:
   `Sum(CostOfGoodsSold)`
9. Set the **Name** to *COGS* and add the following description: *Cost of goods sold at order line level*.
10. Set the **Number formatting** to *Money*.
11. Add four tags: *Cost*, *Line*, *Order*, and *Goods*. Then, click the **Create** button.
12. Navigate into **Analyze** mode and open the **Master Items** sheet.
13. Enable the edit mode and select the table object you created previously.
14. From the **Assets** panel, locate the *Sales* Master Measure. Double-click to replace the existing *Sales* column. Notice that you are not prompted to choose an aggregation function. This is because the Master Measure includes the **Sum** function.
15. Double-click the *COGS* Master Measure to add it to the table object.
16. Disable the edit mode and notice that the table now contains three columns that are all **Master items** in the library. The use of these fields in other objects means they are all based on the same definition.

> **Note:** You can change a **Master Dimension** or **Master Measure** by selecting it from the **Assets** panel, in the **Master items** tab, and clicking **Edit** in the pop-up.
> The changes apply to all the visualizations that use the **Master item**.

Solution



Year	Sales	COGS
Totals	**$12,233,524.28**	**$9,510,792.29**
2020	$269,622.03	$201,489.78
2021	$2,028,051.47	$1,572,566.54
2022	$5,053,637.56	$3,936,846.35
2023	$4,882,213.22	$3,799,889.62

Activity 10 - Optional: Master Measures with set analysis

Summary:

- Create two Master Measures within analysis mode, using set analysis syntax.
- Leverage the two Master Measures inside KPI objects.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, create two new Master Measures, based on more complex expressions, by doing the following:

- From the **Assets** panel, create a new Master Measure and use a set analysis **Expression** to only calculate America's number of orders.
- Use *Orders America* as the Master Measure **Name**.

- Repeat the process to create an *Orders EMEA* Master Measure. Use set analysis for that as well.
- Create two **KPI** objects on the sheet, for *Orders America* and *Orders EMEA* respectively, to show the results.

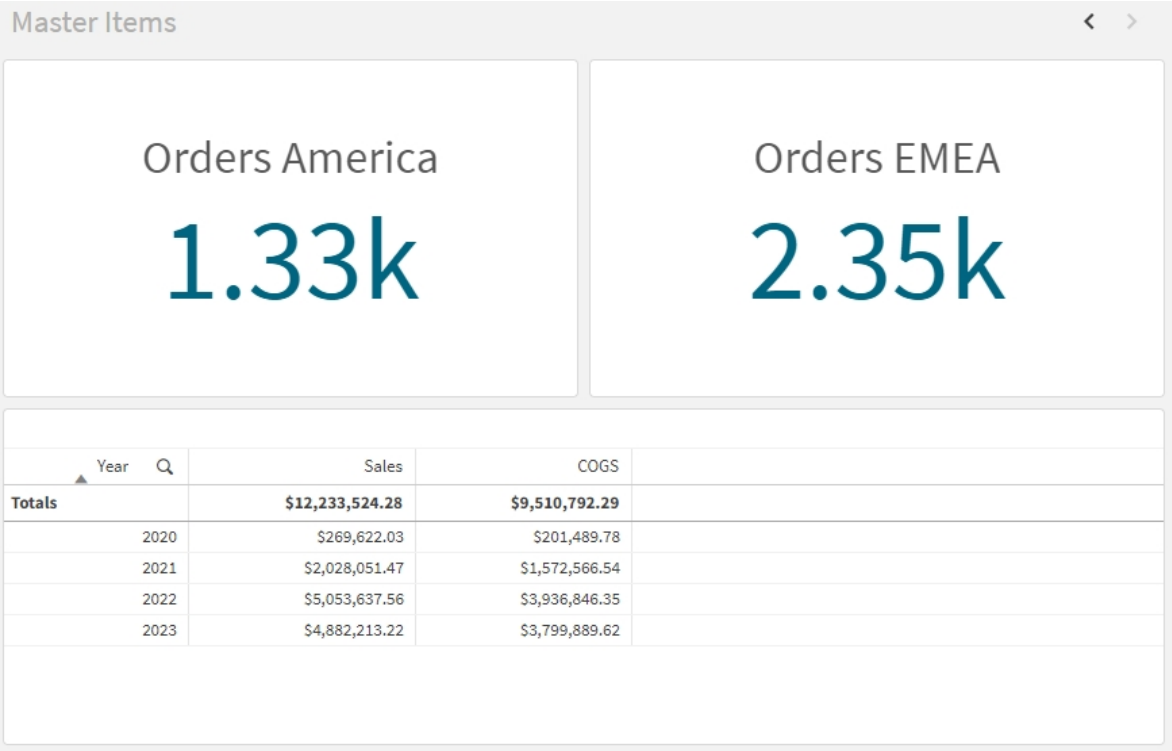
Hints

- The America expression calculates the number of orders for both North and South America divisions.
- **Orders America:**
  `Count({$<DivisionName={"*America*"}>} OrderRecordCounter)`
- **Orders EMEA:**
  `Count({$<DivisionName={'EMEA'}>} OrderRecordCounter)`

Step-by-step guide

1. Edit the **Master Items** sheet. Create a new Master Measure using the following **Expression**:
   `Count({$<DivisionName={"*America*"}>} OrderRecordCounter)`
2. Use *Orders America* as the Master Measure **Name**, and set the **Description** to *Orders in America region*.
3. Add two **Tags**: *Orders* and *America*. Then, click the **Create** button.
4. Create a second Master Measure using the following **Expression**:
   `Count({$<DivisionName={'EMEA'}>} OrderRecordCounter)`
5. Name the Master Measure *Orders EMEA* and set the **Description** to *Orders in EMEA region*.
6. Add two **Tags**: *Orders* and *EMEA*. Then, click the **Create** button.
7. From the **Assets** panel, add two **KPI** objects onto the sheet.
8. Add the *Orders America* Master Measure to the first **KPI** object and the *Orders EMEA* to the second **KPI** object.
9. Disable the edit mode and observe the results.

Solution



Year	Sales	COGS
Totals	**$12,233,524.28**	**$9,510,792.29**
2020	$269,622.03	$201,489.78
2021	$2,028,051.47	$1,572,566.54
2022	$5,053,637.56	$3,936,846.35
2023	$4,882,213.22	$3,799,889.62

Activity 11 - Optional Challenge: More Master items

Summary:

- Generate additional Master Measures and Master Dimensions using complex functions and set analysis.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, add more **Master items** using the methods you have learned. Use the table below to achieve that.

Note: The objective of this **challenge** is to test whether you feel comfortable with the concepts that have been discussed so far in your class. There is no **Step-by-Step** section in this activity, so you must only rely on the **Hints** below to understand the tasks you have to execute.
As you are still new to many of these concepts, please don't hesitate to reach out for more guidance (help file, instructor, Qlik Community).

Hints

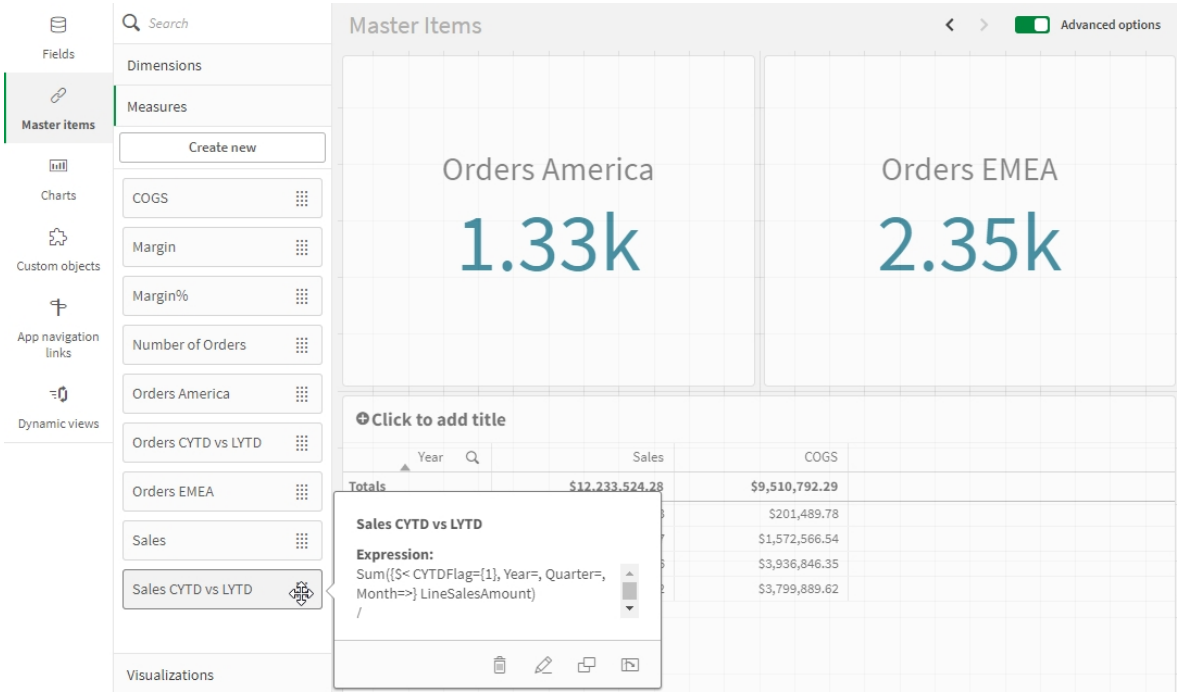
- Review the table below. Add the **Master items** and their expressions in your **ABC Sales Data** app.
- Decide whether each is a dimension or measure to create them.

Master Item	Expression
Number of Orders	`Sum(OrderRecordCounter)`
Margin	`Sum(Margin)`
EMEA Customers	`Aggr(Only({$<DivisionName={EMEA}>} CompanyName), CompanyName)`
Margin%	`Sum(Margin) / Sum(LineSalesAmount)`

Sales CYTD vs LYTD	`Sum({$< CYTDFlag={1}, Year=, Quarter=, Month=>} LineSalesAmount)`
	`/`
	`Sum({$< LYTDFlag={1}, Year=, Quarter=, Month=>} LineSalesAmount)`

Orders CYTD vs LYTD	`Count({$< CYTDFlag={1}, Year=, Quarter=, Month=>} OrderRecordCounter)`
	`/`
	`Count({$< LYTDFlag={1}, Year=, Quarter=, Month=>} OrderRecordCounter)`

Solution



Activity 12: Data islands

Summary:

- Load exchange rate data as a data island in the data model.
- Leverage expressions and field settings to utilize the exchange rate data values within analysis mode.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, include currency exchange data. Do the following to create a multi-currency functionality in the app:

- Load the *ExchangeRates* data set from a local text file.
- Keep the new table as a data island and use expressions in the sheets to display the difference.
- Create a new sheet, *Currencies*, and add the *QuoteCurrency* as a **Filter pane**.
- Insert a **Table** object with:
  - *CompanyName* as the dimension
  - *Sum(LineSalesAmount)* as the first measure with the name *Sales*
  - *Sum(LineSalesAmount) * [ExchangeRate]* as a second measure, with **Number formatting** as *Money* and the following dynamic **Label**:
    = 'Sales in' & Only([QuoteCurrency])
- Change the *QuoteCurrency* properties to only allow one selected value. Select a value and observe the results.

Hints

- The new *ExchangeRate* table is not associated with the existing data model. This means that the association must be created at the expression level.
- The table does not work if more than one value or no values are selected in the *QuoteCurrency* field.

Step-by-step guide

1. In the **Data load editor**, create a new section after **MappingTables** named *Islands*.
2. From the **Data connections** panel, use **Select data** on the **DataFiles** connection.
3. Sourcing the data from the student file collateral, drag the *ExchangeRates.txt* file onto the drop zone to import it.

   Qlik Sense on Windows: Use the **TextFiles** connection you created previously.

4. Once imported, select *ExchangeRates.txt* to open it.

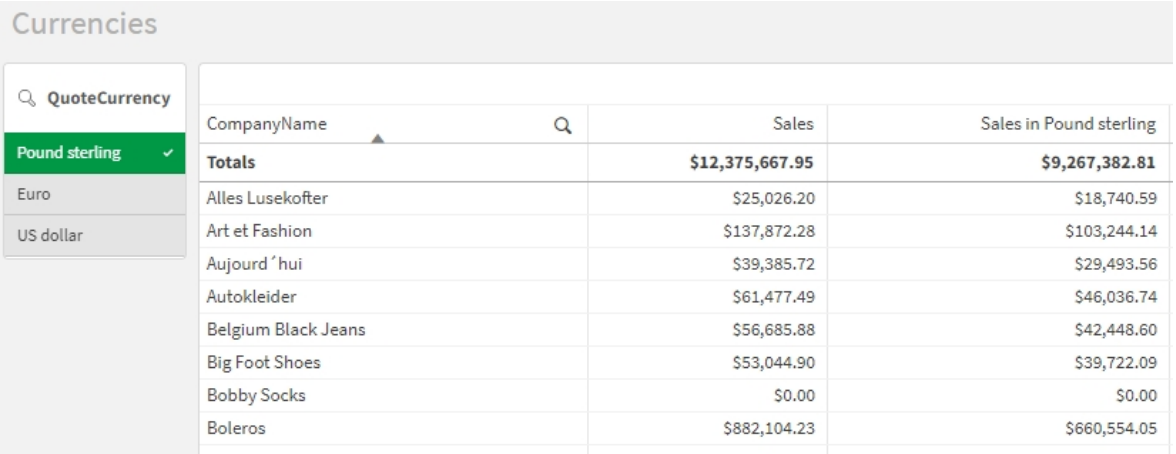
5. Ensure the following settings are being used:
   - **File format**: *Delimited*
   - **Field names**: *Embedded field names*
   - **Delimiter**: *Comma*
   - **Header size**: *0*
   - **Quoting**: *MSQ*
6. Insert the script and label the table *ExchangeRates*. Provide suitable commentary.
7. Update the file source with a variable, if available.
8. Execute the data load script and examine your data model. Note that the new table is a data island, as it is not associated with the rest of the data model.
9. Select the *ExchangeRates* table and expand the **Preview**. This table will be used to convert some measures, like *Sales* amount, into the corresponding selected *QuoteCurrency*.
10. Within **Analyze** mode, create a new sheet, *Currencies*.
11. Add the *QuoteCurrency* field onto the sheet as a **Filter pane**.
12. Insert a **Table** object with *CompanyName* as the dimension and the Master Measure of *Sales* as the first measure.
13. Add a second measure with the following properties:

Label	*= 'Sales in ' & Only([QuoteCurrency])*
Number formatting	*Money*
Measure	`Sum(LineSalesAmount) * [ExchangeRate]`

> **Note:** When using **Money** as the **Number formatting**, Qlik Sense utilizes the settings from the data load script to determine the format pattern. If required, it can be overridden.

14. In order for the above expression to work, only one selected value must be allowed in the *QuoteCurrency* field. From the **Assets** panel, under the **Fields** section, right-click on the *QuoteCurrency* field. Choose **Field settings** and enable the **Always one selected value** check box. This prevents the user from selecting multiple values in the field.
15. Disable the edit mode.
16. Change the currency selected for *Pound sterling* in the *QuoteCurrency* **Filter pane**. Observe the results.
17. Return to edit mode for the *Currencies* sheet.
18. From the **Assets** panel, under the **Fields** section, right-click on the *QuoteCurrency* field again. Choose **Field settings** and disable the **Always one selected value** check box. This returns the field to the normal mode of operation.
19. Disable edit mode and clear all selections.

Solution



CompanyName	Sales	Sales in Pound sterling
Totals	**$12,375,667.95**	**$9,267,382.81**
Alles Lusekofter	$25,026.20	$18,740.59
Art et Fashion	$137,872.28	$103,244.14
Aujourd´hui	$39,385.72	$29,493.56
Autokleider	$61,477.49	$46,036.74
Belgium Black Jeans	$56,685.88	$42,448.60
Big Foot Shoes	$53,044.90	$39,722.09
Bobby Socks	$0.00	$0.00
Boleros	$882,104.23	$660,554.05

Activity 13: Create and Load QVD files

Summary:

- Write data load script to store tables and fields into QVD files.
- Write data load script to load table data from QVD files.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, create .qvd files based on the *MasterCalendar* and *Customers* tables. Achieve that by completing the following tasks:

- Create a new script section before **LoadComplete** and name it *ExportToQVD*.

  Qlik Sense on Windows: Create a new data folder connection, *QVDFiles*, that points to the **QVDFiles** folder in **C:\Qlik Sense Data**.

- Navigate to the **ExportToQVD** script section and create script that stores the *MasterCalendar* table in the *MCal.qvd*.
- Do the same for the *Customers* table storing only the *CustomerID* as *CustID*, *CompanyName* as *CustName* and *DivisionName* as *Division*. Name the file *Cust.qvd*.
- Load the data and check the destination storage location to verify that two new files have been created.
- Comment out the entire script in the **MasterCalendar** section and create a new section right below, named *MasterCalendarQVD.*
- Load the *MCal.qvd* using the data connection.
- At the end of the same section, load the *Cust.qvd* file.

- Load the data again and be prepared to explain why the *QVDCustomers* table is not QVD optimized.
- In the **ExportToQVD** section, create a loop that will export all the loaded tables into .qvd files with the same name.
- Examine the destination storage location to verify that all the tables have been created.

Hints

- The **STORE** command needs to be used to create the .qvd files.
- A **FOR...NEXT** loop can be used to export all the tables into .qvd files.

Step-by-step guide

1. Add a new script section before **LoadComplete** and name it *ExportToQVD*.

   Qlik Sense on Windows: Create a new data folder connection, *QVDFiles*, that points to the **QVDFiles** folder in **C:\Qlik Sense Data**.

2. In the **ExportToQVD** section, add the following **STORE** statements to create the .qvd files:

```
STORE MasterCalendar INTO 'lib://DataFiles/MCal.qvd' (qvd);

STORE
    CustomerID AS CustID,
    CompanyName AS CustName,
    DivisionName AS Division
FROM Customers
INTO 'lib://DataFiles/Cust.qvd' (qvd);
```

   Copy to clipboard

   Qlik Sense on Windows: Use *lib://QVDFiles (qse_student)/* as the connection destination.

3. Load the data and verify that the two files exist in the specified location. Any subsequent data loads overwrite the .qvd files.
4. Comment out the **STORE** statements because the data required is now stored in the .qvd files.
5. In the **MasterCalendar** section, comment out all of the script it contains. (**Tip:** You can use shortcut keys *CTRL+A* and then *CTRL+K*).
6. Create a new section after **MasterCalendar** and name it *MasterCalendarQVD*.
7. Navigate to the **MasterCalendarQVD** script section and use the data connection to load all the fields from *MCal.qvd*.
8. With no table label, the table created from this **LOAD** statement is automatically named *MCal*, after the file name. Add the label, *MasterCalendar*, immediately before the **LOAD** statement.
9. At the end of the same section, use the same data connection again to load all the fields from *Cust.qvd*.

10. Add the label *QVDCustomers*. Add the following **WHERE** clause in the end of the **LOAD** statement, right before the semi-colon (;):

```
WHERE Division = 'EMEA';
```

11. Load the data to add the *MasterCalendar* and *QVDCustomer* tables from the .qvd files to your data model.
12. Examine the messages in the **Data load progress** dialog and consider why the *QVDCustomers* table is not QVD optimized:

```
MasterCalendar << MCal
(QVD (row-based) optimized)
Lines fetched: 1,186
QVDCustomers << Cust
Lines fetched: 55
```

13. Go back to the **ExportToQVD** section and add the following script:

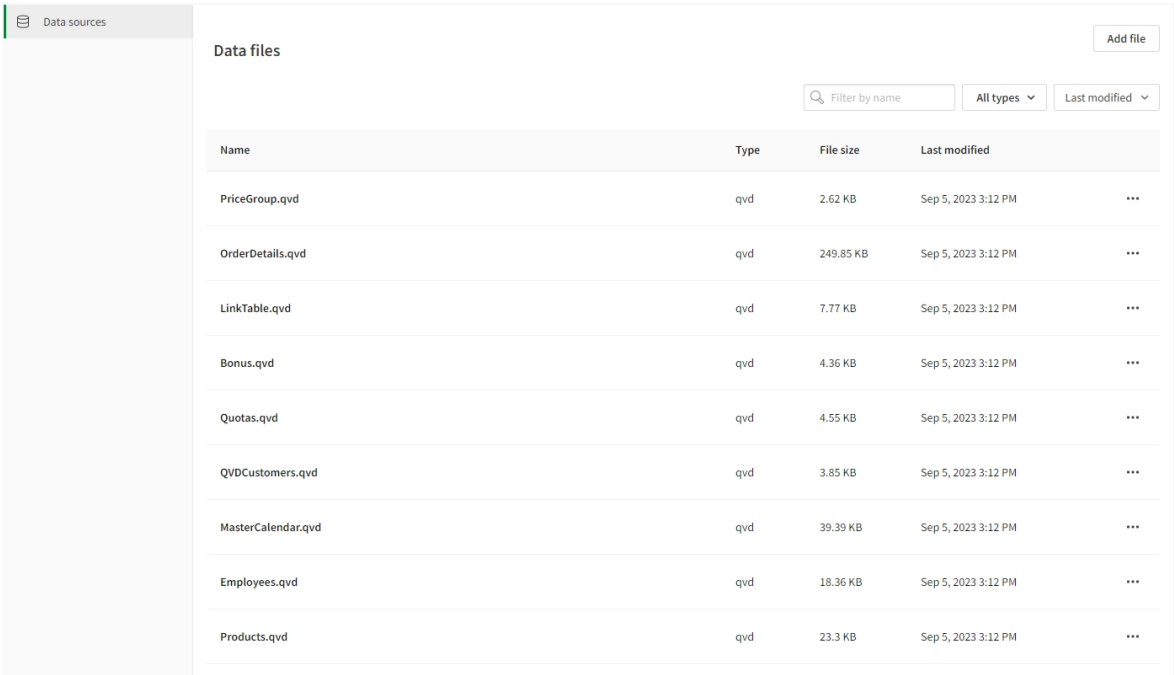
```
FOR i = 0 TO NoOfTables() - 1
    LET d = TableName(i);
    STORE [$(d)] INTO [lib://DataFiles/$(d)].qvd;
NEXT;
```

Copy to clipboard

Qlik Sense on Windows: Use *lib://QVDFiles (qse_student)/* as the connection destination.

14. Load the data and check the destination connection location. Verify that all the tables exist in that storage area as .qvd files.

Solution



Activity 14 - Optional: HidePrefix and Rename

Summary:

- Set up a prefix character for hiding fields from the data model.
- Rename key fields to ensure they are not visible to front-end developers.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, hide the field names that begin with a % symbol by doing the following:

- In the **Main** script section, create the **HidePrefix** variable with the value `'%'`.
- In the **LoadComplete** script section, before the **vLoadComplete** variable, rename the following fields:
  - *OrderID* to *%OrderKey*
  - *EmpYrQtr_Key* to *%EmpYrQtr_Key*
  - *CustomerID* to *%CustomerKey*
  - *EmployeeID* to *%EmployeeKey*
  - *ProductID* to *%ProductKey*
- Open a random app sheet and edit it. In the **Assets** panel, search for *Order*. Verify that the *%OrderKey* does not appear, because it is hidden.
- Search for more fields you have hidden.

Hints

- The **SET** statement can be used to create a variable.
- At the end of the script, a number of **RENAME** statements must be used with the addition of a % symbol at the start of field names, hidden from the users.

Step-by-step guide

1. Select the **Main** script section and add the following line at the end of the **SET** statements:

```
SET HidePrefix = '%';
```

2. Navigate to the **LoadComplete** section and add the following before the *vLoadComplete* variable.

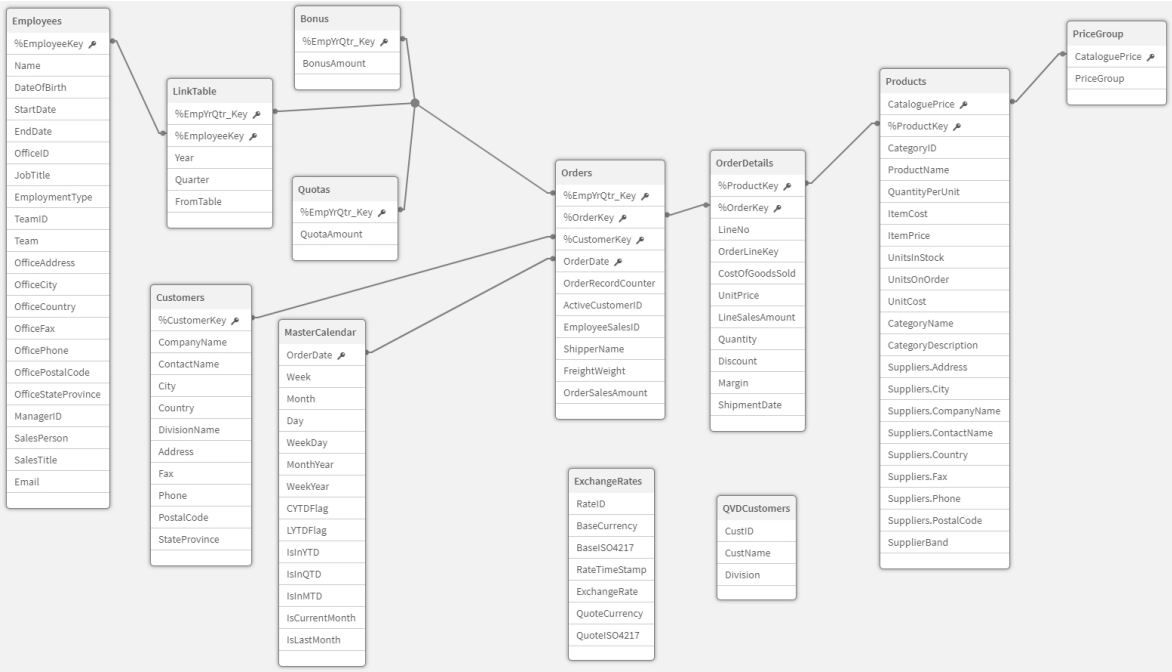
```
// **** Rename key fields to hide ****
RENAME FIELD OrderID TO %OrderKey;
RENAME FIELD EmpYrQtr_Key TO %EmpYrQtr_Key;
RENAME FIELD CustomerID TO %CustomerKey;
```

```
RENAME FIELD EmployeeID TO %EmployeeKey;
RENAME FIELD ProductID TO %ProductKey;
```

Copy to clipboard

3. Load the data, switch to **Analyze** mode, and open a random sheet.
4. Enable the edit mode on the sheet. Navigate to the **Fields** in the **Assets** panel.
5. Search for *Order*. Verify that the list of fields displayed from the search does not contain either *%OrderKey* or *OrderID*, because *OrderID* is renamed and hidden.
6. Search for other fields you have hidden to verify they are not there.

Note: Although you can hide fields from the users, you have to consider if they are needed. *%OrderKey* may be a key field and is hidden, but it does contain *OrderID* information, by which the users may want to search. In this case, you would need to add *OrderID* as a separate non-key field.

Note: As opposed to hiding, it is advisable to **Drop** unused, non-key fields from the data model.

Solution